

MUHAMMAD AKBAR

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EDUCATION

NED University of Engineering and Technology

Bachelor of Science (B.S.) in Computer Science and Information Technology

Karachi, Pakistan

Oct 2022 – Jun 2026 (Expected)

Bahria College Karsaz

Pre-Engineering

Karachi, Pakistan

Aug 2020 – July 2022

EXPERIENCE

Integration Xperts

Data Science Intern

Karachi, Pakistan

June 2025 – August 2025

- Assisted in the development of computer vision solutions, specifically focusing on factory compliance monitoring and a facial recognition-based access system, contributing to the advancement of AI-driven security and safety measures.
- Implemented automated testing using Pytest to rigorously evaluate the performance of age and gender classification models on CCTV image datasets, ensuring high accuracy and reliability in real-world applications.
- Created and managed Regions of Interest (ROIs) for retail use cases, strategically defining areas for focused analysis and integrating these configurations into the project database to enhance data-driven decision-making.

Karachi Development Authority, IT Department

IT Intern

Karachi, Pakistan

March 2025 – April 2025

- Supported IT infrastructure maintenance to ensure system reliability, proactively addressing potential issues and maintaining optimal performance.
- Contributed to database management efforts, ensuring data accuracy and integrity through meticulous attention to detail and adherence to established protocols.
- Collaborated with the IT team on process documentation and system improvements within government IT operations, enhancing efficiency and effectiveness.

PROJECTS

AI-Powered Financial Research Assistant | None

- Architected an agentic hybrid RAG system using FastAPI and LangGraph, combining unstructured semantic search (Pinecone) with structured XBRL financial fact retrieval (PostgreSQL) to deliver accurate and reliable financial insights.
- Developed a semantic routing engine to classify user intent (quantitative vs. qualitative), dynamically grounding numerical queries in structured financial data to mitigate LLM hallucinations and ensure data-driven responses.
- Engineered a concurrent, just-in-time ETL pipeline using AsyncIO that detects missing SEC filings during live queries and triggers parallel scraping, parsing, and indexing (text and XBRL) for real-time responses, demonstrating expertise in efficient data processing.

AI-Powered Factory Compliance System | None

- Developed a real-time computer vision system using YOLOv8 to detect PPE violations, machine occupancy, worker flow anomalies, and cleaning events, enhancing workplace safety and operational efficiency.
- Designed custom Regions of Interest (ROIs) with rule-based compliance tracking for precise, zone-specific monitoring, enabling targeted analysis and improved compliance management.
- Integrated modular detectors (helmet, glove, occupancy, cleaning) within a scalable architecture, logging violations, occupancy durations, and shift-wise summaries for automated compliance reporting, showcasing strong system design and integration skills.

AI-Powered Face Recognition and Attendance System | None

- Developed a real-time facial recognition access system using MTCNN, FaceNet, and DeepSort with embedding-based matching via TensorFlow and DeepFace, improving security and access control.
- Automated frame extraction, image augmentation, and database logging with OpenCV and OpenPyXL for efficient face-based access tracking, optimizing data processing and storage.
- Designed a Streamlit dashboard for live monitoring and user registration, implementing modular OOP architecture with logging and exception handling, demonstrating proficiency in full-stack development and software engineering principles.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, SQL

AI/ML: Computer Vision, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Data Science, Deep Learning

Technical Skills: Object-Oriented Programming, Data Structures & Algorithms, Backend API Development

Other: Problem Solving